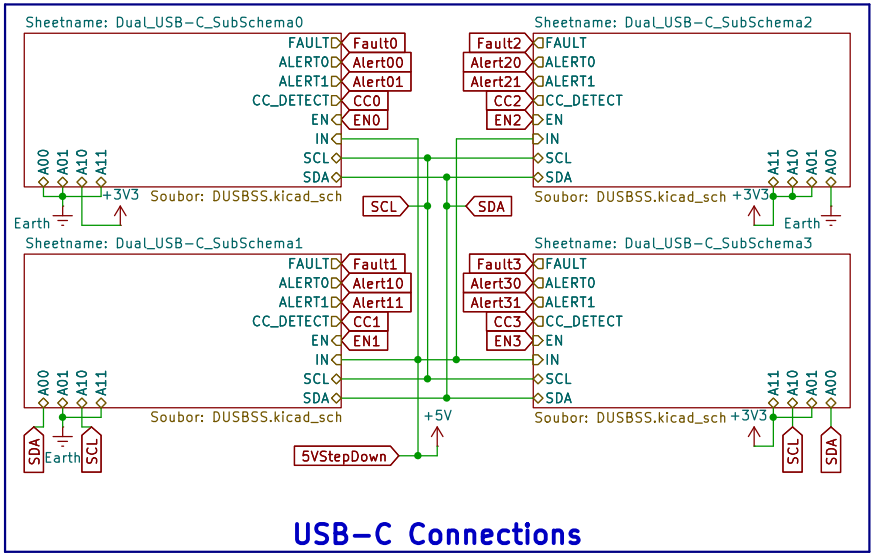
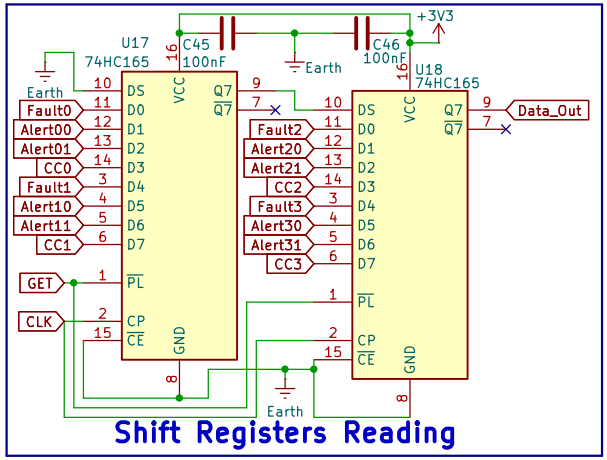
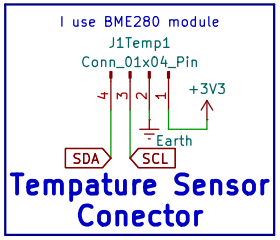
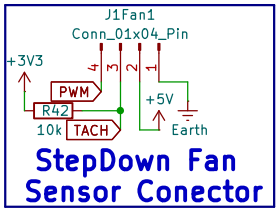
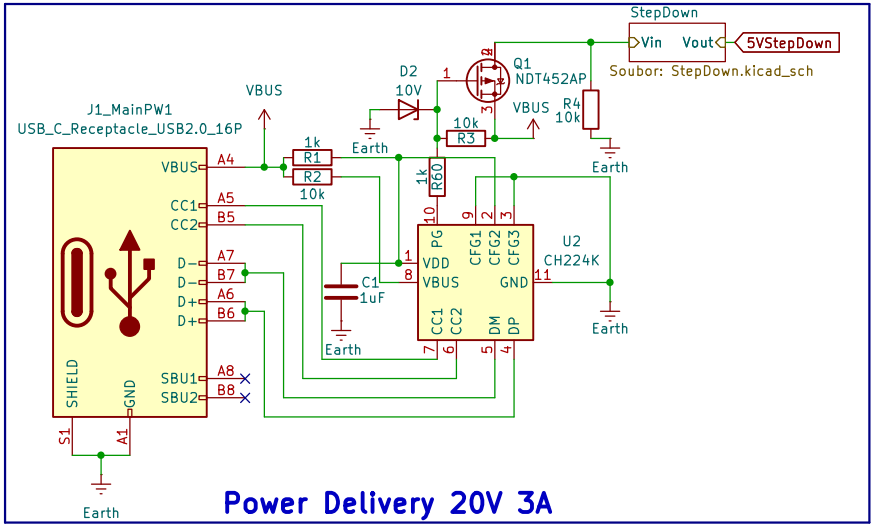
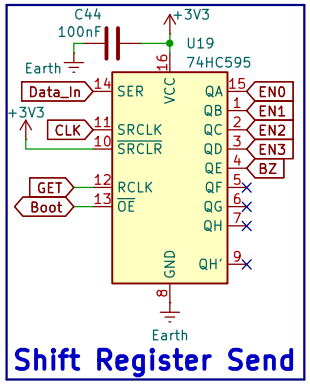

Sheetname: Dual_USB-C_SubSchema0
Sheetname: Dual_USB-C_SubSchema2
Sheetname: Dual_USB-C_SubSchema1
Sheetname: Dual_USB-C_SubSchema3
Soubor: DUSBSS.kicad_sch
FAULT
ALERT0
ALERT1
CC_DETECT
EN
IN
SCL
SDA
Fault0
Alert00
Alert01
CC0
EN0
Fault2
Alert20
Alert21
CC2
EN2
Fault1
Alert10
Alert11
CC1
EN1
Fault3
Alert30
Alert31
CC3
EN3
A00
A01
A10
A11
+3V3
+5V
Earth
5VStepDown
USB-C Connections

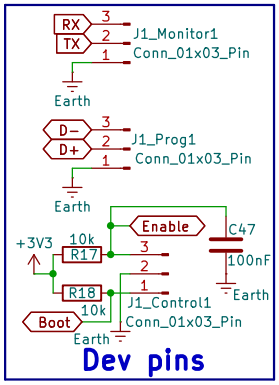

U17
74HC165
U18
74HC165
C45
100nF
C46
100nF
+3V3
Earth
Fault0
Alert00
Alert01
CC0
Fault1
Alert10
Alert11
CC1
Fault2
Alert20
Alert21
CC2
Fault3
Alert30
Alert31
CC3
GET
CLK
Data_Out
DS
D0
D1
D2
D3
D4
D5
D6
D7
PL
CP
CE
VCC
GND
Q7
Shift Registers Reading

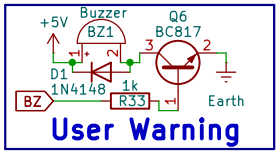

I use BME280 module
J1Temp1
Conn_01x04_Pin
+3V3
Earth
SDA
SCL
Tempature Sensor Conector

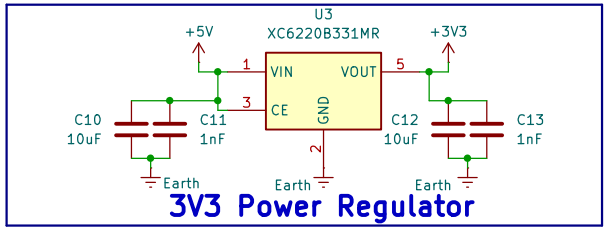

J1Fan1
Conn_01x04_Pin
+3V3
+5V
PWM
TACH
R42
10k
Earth
StepDown Fan Sensor Conector

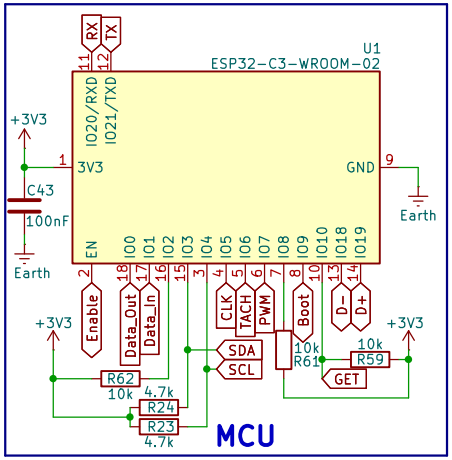

StepDown
Vin
Vout
5VStepDown
Soubor: StepDown.kicad_sch
J1_MainPW1
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D+
SHIELD
GND
SBU1
SBU2
D2
10V
Q1
NDT452AP
R1
R2
R3
R4
R60
10k
1k
C1
1uF
U2
CH224K
PG
VDD
CFG1
CFG2
CFG3
DM
DP
Earth
Power Delivery 20V 3A

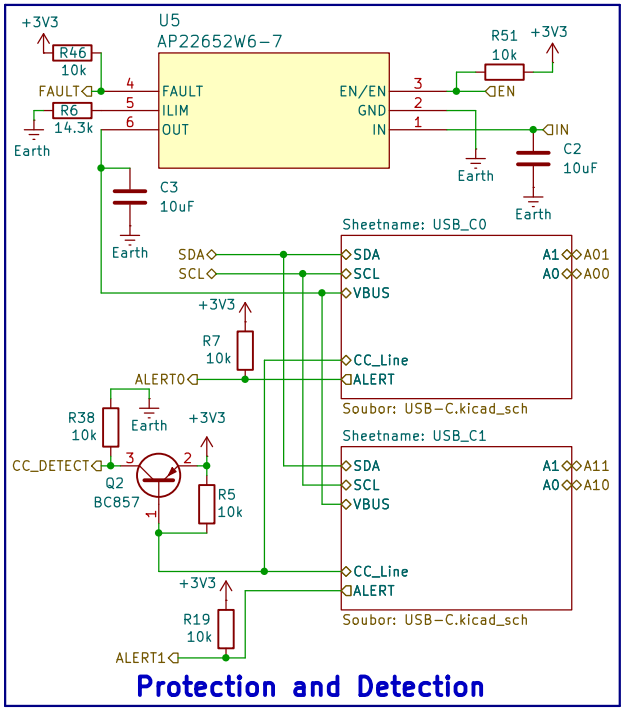

C44
100nF
U19
74HC595
+3V3
Earth
Data_In
CLK
GET
Boot
SER
SRCLK
SRCLR
RCLK
OE
QA
QB
QC
QD
QE
QF
QG
QH
QH'
GND
VCC
EN0
EN1
EN2
EN3
BZ
Shift Register Send

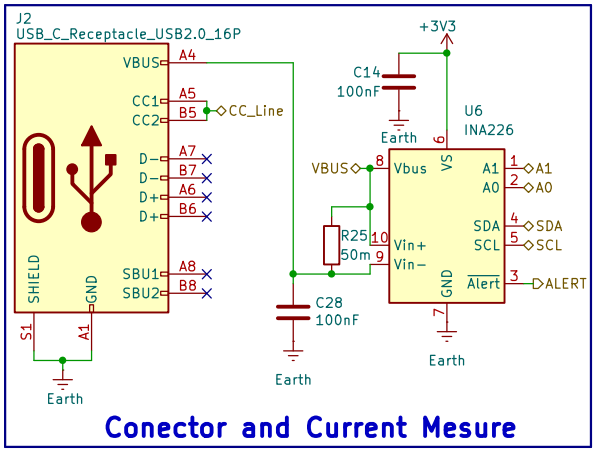

RX
TX
J1_Monitor1
Conn_01x03_Pin
D-
D+
J1_Prog1
J1_Control1
Enable
Boot
R17
R18
10k
C47
100nF
Earth
+3V3
Dev pins

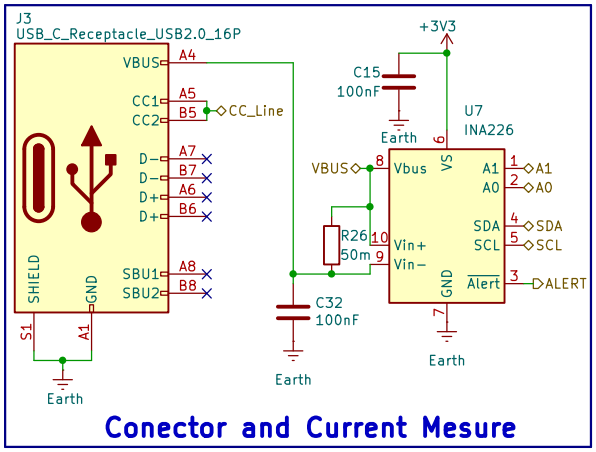

+5V
Buzzer
BZ1
Q6
BC817
D1
1N4148
R33
1k
BZ
Earth
User Warning

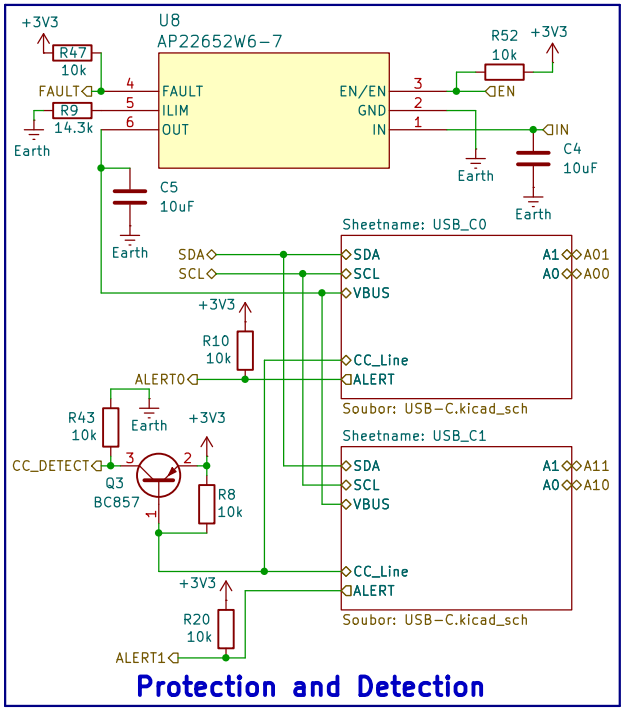

U3
XC6220B331MR
+5V
+3V3
VIN
VOUT
CE
GND
C10
10uF
C11
1nF
C12
10uF
C13
1nF
Earth
3V3 Power Regulator

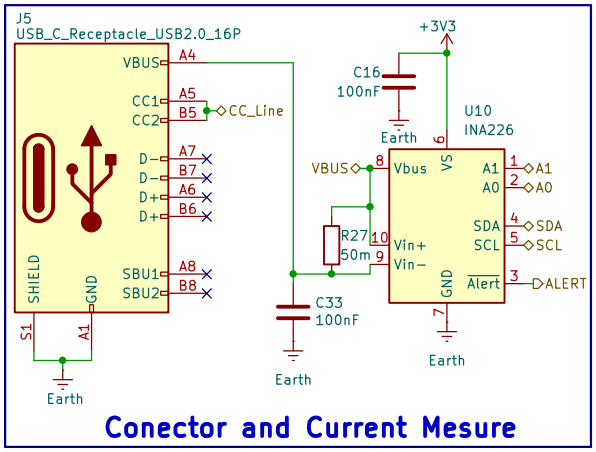

U1
ESP32-C3-WROOM-02
RX
TX
IO20/RXD
IO21/TXD
3V3
GND
EN
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO18
IO19
C43
100nF
Enable
Data_Out
Data_In
CLK
TACH
PWM
Boot
D-
D+
SDA
SCL
R61
10k
R59
10k
GET
R62
10k
R24
4.7k
R23
4.7k
+3V3
Earth
MCU

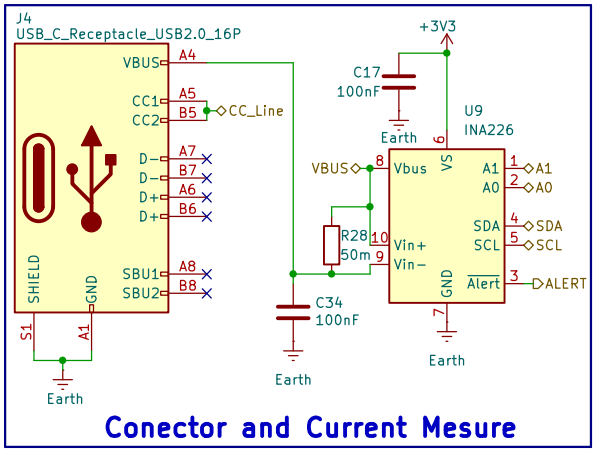

+3V3
U5
AP22652W6-7
R46
10k
FAULT
4
5
6
R6
14.3k
Earth
FAULT
ILIM
OUT
EN/EN
GND
IN
3
2
1
R51
10k
+3V3
EN
IN
C2
10uF
Earth
Earth
C3
10uF
Earth
Sheetname: USB_C0
SDA
SCL
VBUS
CC_Line
ALERT
A1
A0
A01
A00
Soubor: USB-C.kicad_sch
+3V3
R7
10k
ALERT0
R38
10k
Earth
+3V3
CC_DETECT
3
2
Q2
BC857
1
R5
10k
Sheetname: USB_C1
SDA
SCL
VBUS
CC_Line
ALERT
A1
A0
A11
A10
Soubor: USB-C.kicad_sch
+3V3
R19
10k
ALERT1
Protection and Detection

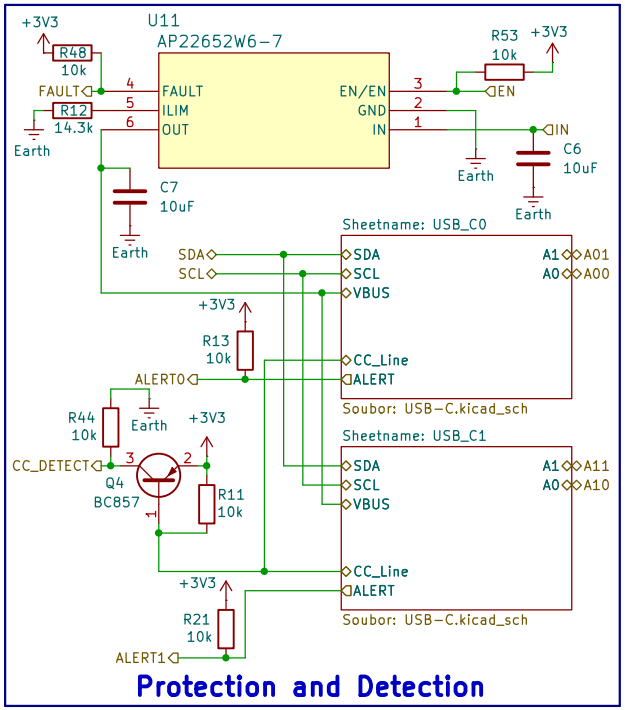

J2
USB_C_Receptacle_USB2.0_16P
VBUS
A4
CC1
A5
CC2
B5
CC_Line
D−
A7
D−
B7
D+
A6
D+
B6
SBU1
A8
SBU2
B8
SHIELD
GND
S1
A1
Earth
C28
100nF
Earth
+3V3
C14
100nF
Earth
U6
INA226
VBUS
8
Vbus
VS
6
R25
50m
10
Vin+
9
Vin−
GND
7
Earth
A1
1
A1
A0
2
A0
SDA
4
SDA
SCL
5
SCL
Alert
3
ALERT
Conector and Current Mesure

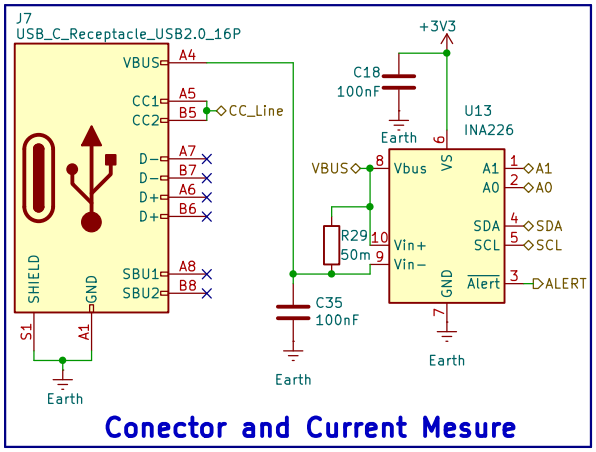

J3
USB_C_Receptacle_USB2.0_16P
VBUS
A4
CC1
A5
CC2
B5
CC_Line
D−
A7
D−
B7
D+
A6
D+
B6
SBU1
A8
SBU2
B8
SHIELD
GND
S1
A1
Earth
C32
100nF
Earth
+3V3
C15
100nF
Earth
U7
INA226
VBUS
8
Vbus
VS
6
R26
50m
10
Vin+
9
Vin−
GND
7
Earth
A1
1
A1
A0
2
A0
SDA
4
SDA
SCL
5
SCL
Alert
3
ALERT
Conector and Current Mesure

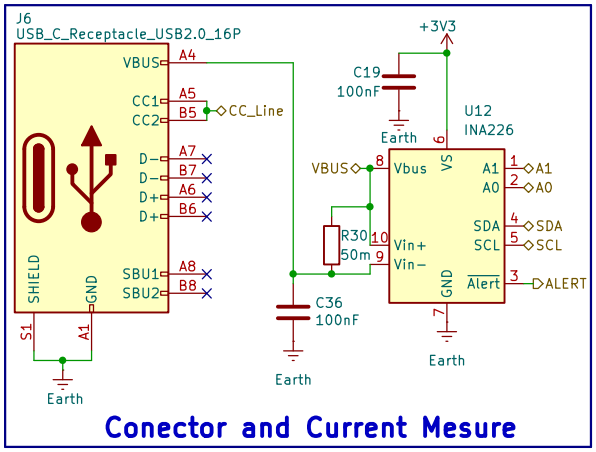

U8
AP22652W6-7
+3V3
R47
10k
FAULT
R9
14.3k
Earth
4
5
6
FAULT
ILIM
OUT
EN/EN
GND
IN
3
2
1
R52
10k
+3V3
EN
IN
Earth
C4
10uF
Earth
C5
10uF
Earth
Sheetname: USB_C0
SDA
SCL
VBUS
CC_Line
ALERT
A1
A0
A01
A00
Soubor: USB-C.kicad_sch
+3V3
R10
10k
ALERT0
R43
10k
Earth
+3V3
CC_DETECT
3
2
1
Q3
BC857
R8
10k
Sheetname: USB_C1
SDA
SCL
VBUS
CC_Line
ALERT
A1
A0
A11
A10
Soubor: USB-C.kicad_sch
+3V3
R20
10k
ALERT1
Protection and Detection

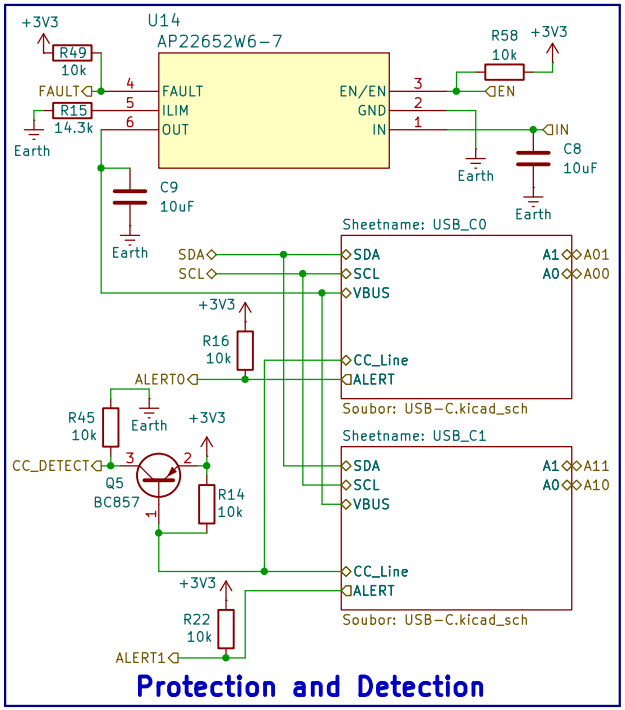

J5
USB_C_Receptacle_USB2.0_16P
VBUS
A4
CC1
A5
CC2
B5
CC_Line
D−
A7
D−
B7
D+
A6
D+
B6
SBU1
A8
SBU2
B8
SHIELD
GND
S1
A1
Earth
+3V3
C16
100nF
Earth
U10
INA226
VBUS
8
Vbus
VS
6
A1
1
A1
A0
2
A0
SDA
4
SDA
SCL
5
SCL
R27
50m
10
Vin+
9
Vin−
GND
7
Alert
3
ALERT
C33
100nF
Earth
Earth
Conector and Current Mesure

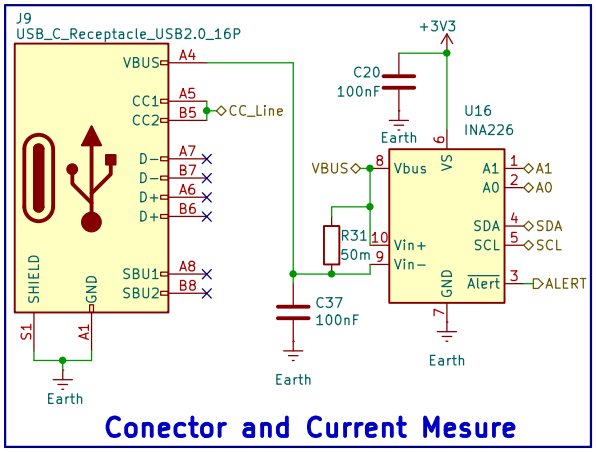

J4
USB_C_Receptacle_USB2.0_16P
VBUS
A4
CC1
A5
CC2
B5
CC_Line
D−
A7
D−
B7
D+
A6
D+
B6
SBU1
A8
SBU2
B8
SHIELD
GND
S1
A1
Earth
+3V3
C17
100nF
Earth
U9
INA226
VBUS
8
Vbus
VS
6
R28
50m
10
Vin+
9
Vin−
GND
7
Earth
A1
1
A1
A0
2
A0
SDA
4
SDA
SCL
5
SCL
Alert
3
ALERT
C34
100nF
Earth
Conector and Current Mesure

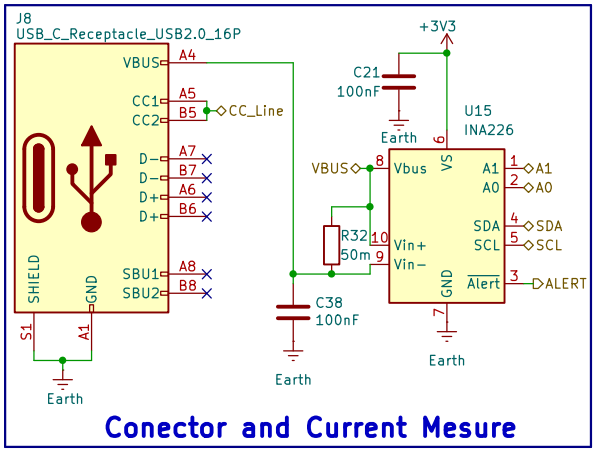

+3V3
U11
AP22652W6-7
R48
10k
FAULT
4
5
6
R12
14.3k
Earth
FAULT
ILIM
OUT
EN/EN
GND
IN
3
2
1
R53
10k
+3V3
EN
IN
C6
10uF
Earth
Earth
C7
10uF
Earth
Sheetname: USB_C0
SDA
SCL
VBUS
CC_Line
ALERT
A1
A0
A01
A00
Soubor: USB-C.kicad_sch
+3V3
R13
10k
ALERT0
R44
10k
Earth
+3V3
CC_DETECT
3
2
1
Q4
BC857
R11
10k
Sheetname: USB_C1
SDA
SCL
VBUS
CC_Line
ALERT
A1
A0
A11
A10
Soubor: USB-C.kicad_sch
+3V3
R21
10k
ALERT1
Protection and Detection

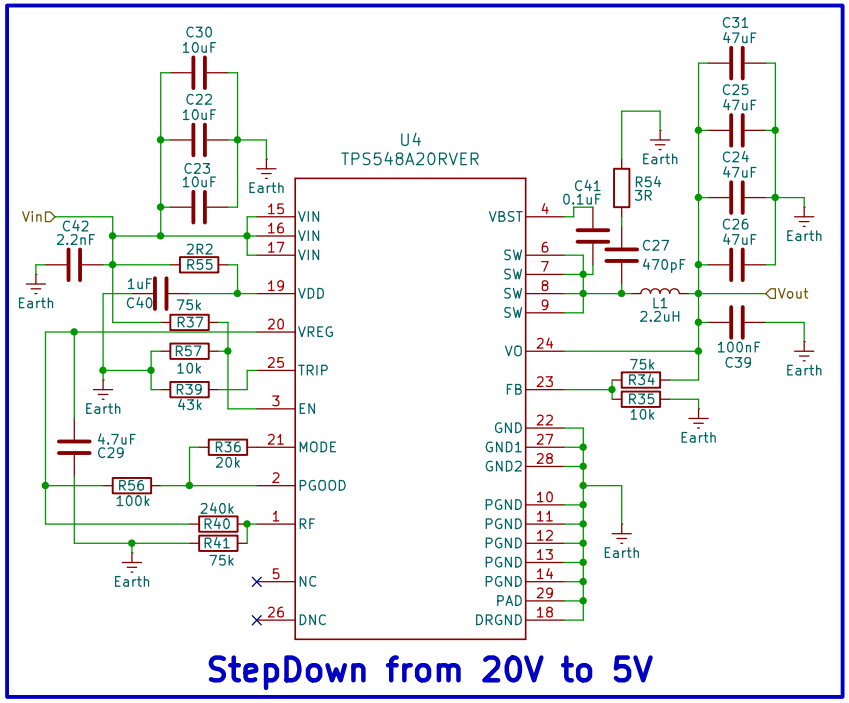

J7
USB_C_Receptacle_USB2.0_16P
VBUS
A4
CC1
A5
CC2
B5
CC_Line
D−
A7
D−
B7
D+
A6
D+
B6
SBU1
A8
SBU2
B8
SHIELD
GND
S1
A1
Earth
+3V3
C18
100nF
Earth
U13
INA226
VBUS
8
Vbus
VS
6
R29
50m
10
Vin+
9
Vin−
GND
7
A1
1
A0
2
SDA
4
SCL
5
Alert
3
A1
A0
SDA
SCL
ALERT
C35
100nF
Earth
Earth
Conector and Current Mesure


J6
USB_C_Receptacle_USB2.0_16P
VBUS
A4
CC1
A5
CC2
B5
CC_Line
D−
A7
D−
B7
D+
A6
D+
B6
SBU1
A8
SBU2
B8
SHIELD
GND
S1
A1
Earth
C19
100nF
Earth
+3V3
U12
INA226
VBUS
8
Vbus
VS
6
A1
A0
SDA
SCL
Alert
GND
7
1
2
4
5
3
A1
A0
SDA
SCL
ALERT
R30
50m
10
Vin+
9
Vin−
C36
100nF
Earth
Earth
Conector and Current Mesure


+3V3
U14
AP22652W6-7
R49
10k
FAULT
4
5
6
R15
14.3k
Earth
FAULT
ILIM
OUT
EN/EN
GND
IN
3
2
1
R58
10k
+3V3
EN
IN
Earth
C8
10uF
Earth
C9
10uF
Earth
Sheetname: USB_C0
SDA
SCL
VBUS
CC_Line
ALERT
A1
A0
A01
A00
Soubor: USB-C.kicad_sch
+3V3
R16
10k
ALERT0
R45
10k
Earth
+3V3
CC_DETECT
3
2
1
Q5
BC857
R14
10k
Sheetname: USB_C1
SDA
SCL
VBUS
CC_Line
ALERT
A1
A0
A11
A10
Soubor: USB-C.kicad_sch
+3V3
R22
10k
ALERT1
Protection and Detection


J9
USB_C_Receptacle_USB2.0_16P
VBUS
A4
CC1
A5
CC2
B5
CC_Line
D−
A7
D−
B7
D+
A6
D+
B6
SBU1
A8
SBU2
B8
SHIELD
GND
S1
A1
Earth
+3V3
C20
100nF
Earth
U16
INA226
VBUS
8
Vbus
VS
6
R31
50m
10
Vin+
9
Vin−
GND
7
A1
1
A1
A0
2
A0
SDA
4
SDA
SCL
5
SCL
Alert
3
ALERT
C37
100nF
Earth
Earth
Conector and Current Mesure


J8
USB_C_Receptacle_USB2.0_16P
VBUS
A4
CC1
A5
CC2
B5
CC_Line
D−
A7
D−
B7
D+
A6
D+
B6
SBU1
A8
SBU2
B8
SHIELD
GND
S1
A1
Earth
+3V3
C21
100nF
Earth
U15
INA226
VBUS
8
Vbus
VS
6
A1
A0
SDA
SCL
Alert
ALERT
R32
50m
10
Vin+
9
Vin−
GND
7
C38
100nF
Earth
Earth
Conector and Current Mesure


U4
TPS548A20RVER
C30
10uF
C22
10uF
C23
10uF
Earth
Vin
C42
2.2nF
Earth
2R2
R55
1uF
C40
75k
R37
R57
10k
R39
43k
Earth
4.7uF
C29
R56
100k
R36
20k
240k
R40
R41
75k
Earth
15 VIN
16 VIN
17 VIN
19 VDD
20 VREG
25 TRIP
3 EN
21 MODE
2 PGOOD
1 RF
5 NC
26 DNC
VBST 4
SW 6
SW 7
SW 8
SW 9
VO 24
FB 23
GND 22
GND1 27
GND2 28
PGND 10
PGND 11
PGND 12
PGND 13
PGND 14
PAD 29
DRGND 18
C41
0.1uF
Earth
R54
3R
C27
470pF
L1
2.2uH
C31
47uF
C25
47uF
C24
47uF
C26
47uF
Earth
Vout
100nF
C39
Earth
75k
R34
R35
10k
Earth
Earth
StepDown from 20V to 5V